

Proposal: Replace the sourcing reverse block scan with an index lookup

Status	Proposal
Area	JetStream stream sourcing (<code>server/stream.go</code>) + store index (<code>server/filestore.go</code> , <code>server/memstore.go</code>)
Date	2026-06-06
Companion	<code>sourcing-durable-resume.md</code> (lifecycle, durable consumer, triggers)
Target	<code>startingSequenceForSources</code> (<code>stream.go:4694</code>) and <code>setStartingSequenceForSources</code> (<code>stream.go:4566</code>)

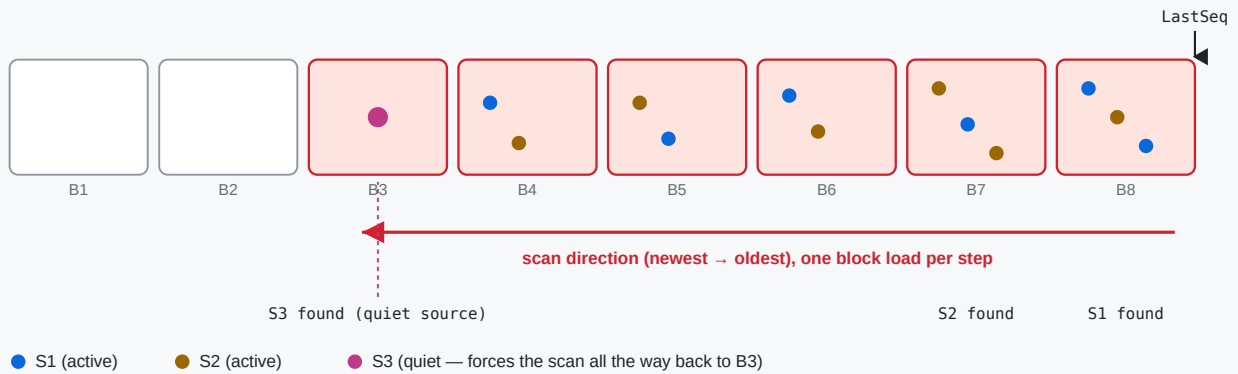
1. The problem in one paragraph

On hub-stream **leader election / restart**, `setupSourceConsumers` calls `startingSequenceForSources` **unconditionally** (`stream.go:4821`). That function reconstructs each source's last sourced sequence by walking the stream's **own store backwards from `LastSeq`**, one block at a time (`LoadPrevMsgMulti` , `filestore.go:9425`), reading the `Nats-Stream-Source` header off matching messages, until **every** source has been located.

Note: the walk is not as naive as message-by-message — within each block the per-block index (`fss`) skips non-matching messages, and the per-source sublist narrows as sources are found. But `LoadPrevMsgMulti` still **visits every block** from `LastSeq` back to the match (loading/decompressing cold ones), and the outer loop **re-walks** as the sublist narrows. So the cost grows with both the number of blocks back to the least-recently-active source (worst case the **entire store**) and the number of sources — all under `mset.mu` .

The reverse block scan: rebuild each source's last sequence from stored headers

startingSequenceForSources walks back from LastSeq, LoadPrevMsgMulti loads (+decompresses) a block at a time until every source is found.



Why it can be slow on a limits-based stream

A single quiet/sparse source (S3) forces the scan past every newer block — worst case the entire stream. Each uncached block is read from disk, decrypted and decompressed. With many sources and a large store, leader election stalls under mset.mu while this runs.

2. The key insight

The store **already maintains a per-subject index** that can return the last sequence for a subject *without* walking messages:

- `fs.psim` — per-subject info, including `lblk` = the **last block index** that contains each subject (`filestore.go:3843-3852` , `8980-8984`).
- `mb.fss` — per-block subject state with `Last` per subject (`filestore.go:9003-9015`).

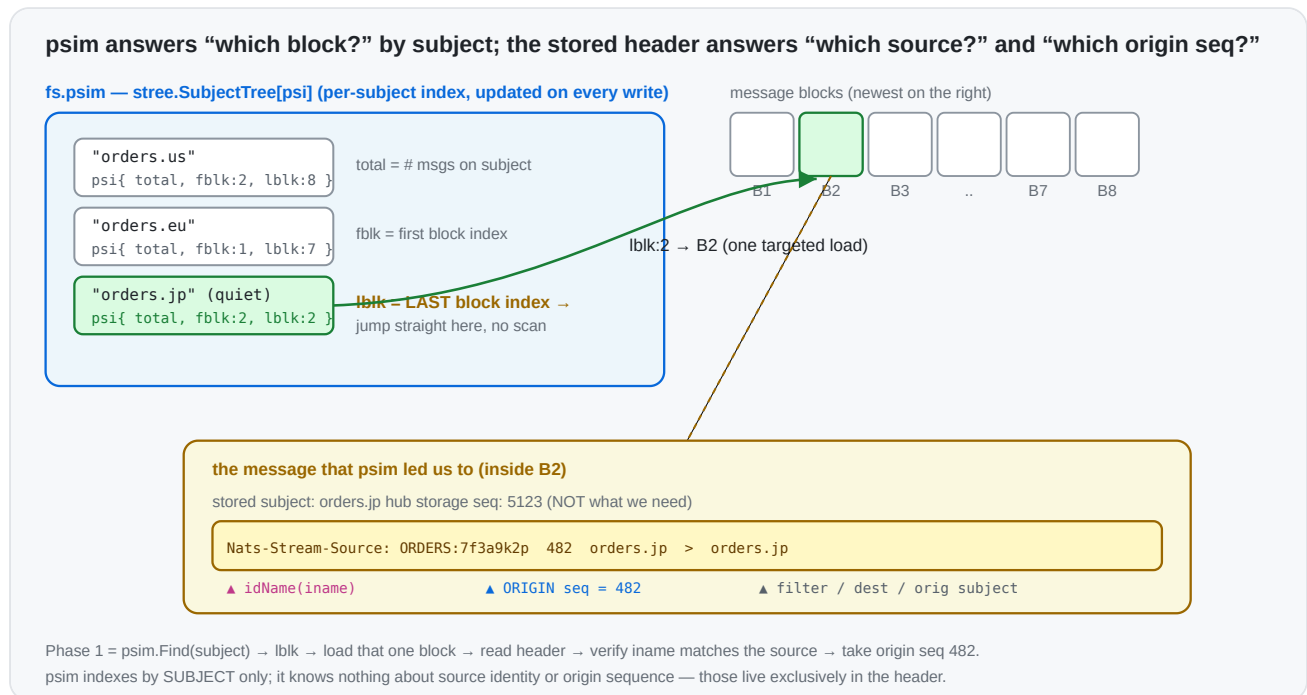
This is what powers `LoadLastMsg(subject) → loadLast ( filestore.go:8939 )`: for a literal subject it jumps straight to `info.lblk` and reads that subject's last message — **one targeted block load**, regardless of how far back it is. `MultiLastSeqs ( filestore.go:3806 )` does the same for a batch of filters in a single index pass.

The current scan ignores this index and instead does a generic backward message walk. The header it needs (the **origin** sequence) is right there in the message that `LoadLastMsg` already returns.

Why a plain "last seq for subject" isn't enough on its own: `si.sseq` must be the **origin** stream sequence taken from the `Nats-Stream-Source` header, not the hub's storage sequence. So we use the index to *find* the source's last stored message in $O(1)$, then read the origin seq from its header.

3. Background: the subject index (`psim` + `fss`) and the source header

The filestore keeps a **two-level subject index**. Phase 1 stands entirely on it, so it's worth being precise about what each level is.



3.1 `psim` — the store-wide "which blocks hold this subject" map

`fs.psim` (`filestore.go:195`) is a `stree.SubjectTree[psi]` — a subject-token tree keyed by the full message subject — whose value is a tiny per-subject record (`filestore.go:168`):

```
type psi struct {
    total uint64 // how many messages exist for this subject across the whole s
    fblk  uint32 // index of the FIRST block that still contains this subject
    lblk  uint32 // index of the LAST block that still contains this subject
}
```

It is maintained incrementally on every write (`filestore.go:4933-4943`): on store, the subject's entry is created (`total:1, fblk=lblk=current block`) or its `total` is bumped and `lblk` advanced to the current block; on removal/expiry the counts and `fblk` / `lblk` are walked forward as blocks empty out. Because `psim` is a *subject tree*, it supports both exact `Find(subject)` and wildcard `Match(filter)` in time proportional to the subject token structure — **not** to the number of messages.

The single field that makes Phase 1 cheap is `lblk`: given a source's subject, `psim.Find` returns the **exact last block** that contains it. There is no need to scan newer blocks — most of which don't contain that subject at all. (`MultiLastSeqs` , `filestore.go:3806` , uses the same `psim` to answer a *batch* of filters in one index pass.)

3.2 `fss` — the per-block "first/last seq of this subject in this block" map

`psim` gets us to a block; `mb.fss` (`filestore.go:239`) finishes the job *inside* that block. Each `msgBlock` carries its own `stree.SubjectTree[SimpleState]` mapping subject → (`store.go:180`):

```
type SimpleState struct {
    Msgs  uint64 // messages for this subject in THIS block
    First uint64 // first sequence for this subject in this block
    Last  uint64 // last  sequence for this subject in this block
    // (firstNeedsUpdate / lastNeedsUpdate: lazily recomputed when interior dele
}
```

So the two levels compose into an O(1)-ish lookup, which is exactly what `loadLast` (`filestore.go:8980-9034`) does:

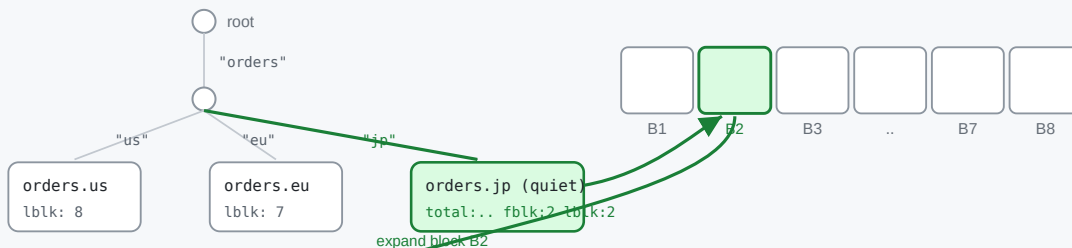
```
psim.Find(subject) → info.lblk → block → block.fss.Find(subject) → ss.Last →
```

`fss` is what gives us the precise sequence within the block (and it correctly skips interior-deleted messages via `lastNeedsUpdate` / `recalculateForSubj`). On a cold block, `ensurePerSubjectInfoLoaded` loads/derives `fss` for that **one** block — the cost we pay once per resolved source, versus the current scan paying it for every block back to the oldest source.

Two-level traversal: psim subject tree → the block → that block's fss subject tree → the message

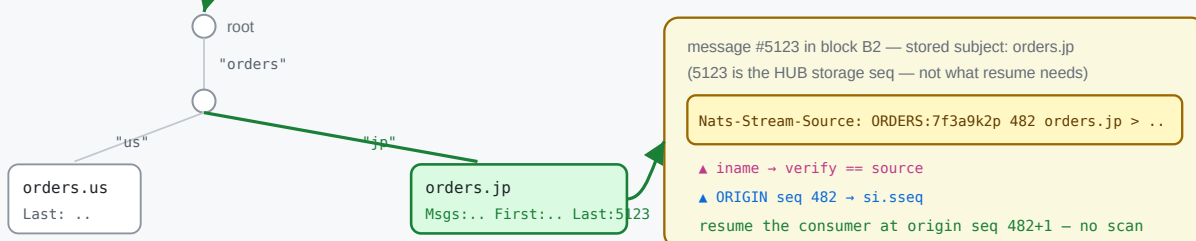
1 fs.psim — store-wide subject tree (subject → psi{total, fblk, lblk})

2 follow lblk → the last block holding that subject



3 B2.fss — per-block subject tree (subject → SimpleState{Msgs, First, Last})

4 load msg at Last (hub seq 5123) → read its header



3.3 Why each message carries a `Nats-Stream-Source` header

psim / fss index by **subject**. They know nothing about *which* source produced a stored message or *where in the origin stream* it came from — and those are precisely the two facts resume needs. That information is carried only in the per-message header `Nats-Stream-Source` (`JSStreamSource`, `stream.go:635`), written by `genSourceHeader` (`stream.go:4470`) as:

```
Nats-Stream-Source: <idName> <originSeq> <filter> <destination> <origSubject>
                    |           |
                    |           └ the ORIGIN stream's sequence – what we must
                    └ origin stream name (+ domain/consumer hash); with filter
                        reconstructs the source's full iname (parsed by stream)
```

This header is needed for three independent reasons:

1. **Source attribution.** A hub message is stored under its *destination* subject; the storage layer records nothing about its origin. When two sources can land on the same/overlapping subject, only the header's `iname` says which source a given message belongs to — hence Phase 1's `iname == si.iname` check.
2. **Origin-sequence tracking.** The sourcing consumer is (re)created on the **origin** with `DeliverByStartSequence`, which lives in the *origin*'s sequence space. The hub's own storage sequence (e.g. 5123) is unrelated to the origin sequence (e.g. 482). Only the

header preserves the origin sequence, so it is the *only* place a resume point can be recovered from without contacting the origin.

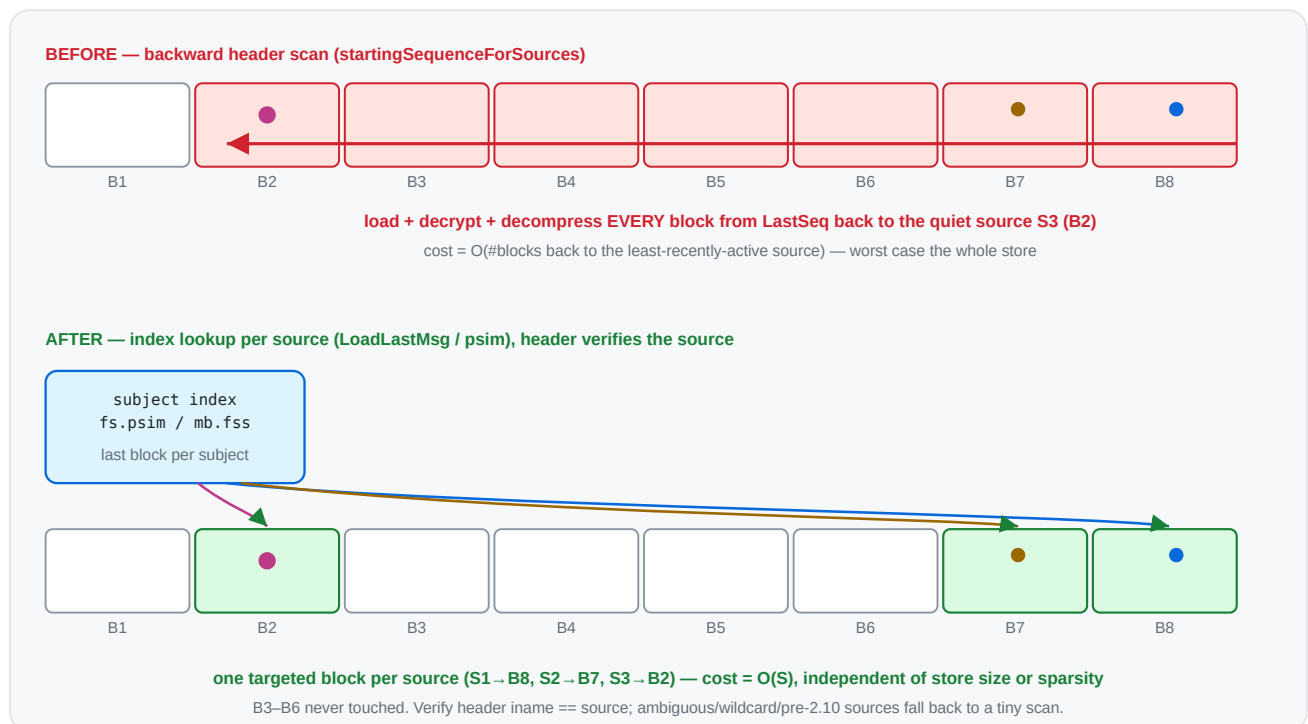
3. Loop / daisy-chain handling & dedup. When re-sourcing ($A \rightarrow B \rightarrow C$),

`processInboundSourceMsg` strips the inbound `JSSource` and stamps its own (`stream.go:4398`, `4405`), so each hop's provenance is well-defined and re-sourced duplicates can be recognised.

This is the crux of the whole proposal: **the index tells us *which block* in $O(1)$; the header (read from that one message) tells us *which source* and *which origin seq***. Today's scan reads the header off *every* message on the way back because it never consults `psim / fss` to jump directly to the right one.

4. Proposed algorithm

Two phases. Phase 1 resolves the common case with index lookups; Phase 2 keeps today's scan only for sources that genuinely can't be attributed by subject.



For each source, its **stored subject** is the `FilterSubject` for a plain source, or — for a source with a **single, concrete (non-templated)** subject transform — the transform **Destination**.

Templated transform destinations contain `{{...}}` mapping tokens (the rendered subject varies per message), and partial wildcards in a destination are rejected at config time, so a destination is concrete unless it is empty, `>`, or contains `{{`.

Phase 1 – index lookup (per source, $O(1)$ targeted load):
for each source `si`:

```

    subj := si.FilterSubject // plain source
    if si has exactly one transform: subj := transform.Destination
    if subj == "" || subj == ">" || subj contains "{": defer to Phase 2; continue
    sm := store.LoadLastMsg(subj) // jumps via psim, ~1 block
    if sm has a JSStreamSource header:
        (_, iname, sseq) := streamAndSeq(header)
        if iname == si.iname: // self-verifying attribution
            si.sseq = sseq; mark resolved
    // anything not resolved (no header / header for a different source / wildcard
    // multi-or-templated transform / pre-2.10) → Phase 2

```

Phase 2 – fallback reverse scan (only for the unresolved subset):

```

    run today's LoadPrevMsgMulti loop, but with the sublist built from ONLY the
    unresolved sources, so it narrows immediately and stops as soon as they are found

```

The header check makes Phase 1 **self-correcting**: if a subject is shared between sources, or overlaps a direct publish on the hub, or the last message has no source header, the lookup simply doesn't match and that source drops to Phase 2 — never producing a wrong answer.

5. Before → after (code)

Before (`startingSequenceForSources` , condensed — `stream.go:4694`)

```

mset.resetSourceInfo()
// build a sublist of ALL sources' subjects
refreshSublist() // every source
for last := state.LastSeq; ; { // walk the WHOLE store backwards
    sm, seq, err := mset.store.LoadPrevMsgMulti(sl, last, &smv) // loads/decompresses
    if err == ErrStoreEOF || err != nil { break }
    last = seq - 1
    ...
    _, iName, sseq := streamAndSeq(bytesToString(ss))
    update(iName, sseq) // narrows sublist as sources are found
    if len(seqs) == expected { return } // stop only when ALL sources are found
}

```

Cost: O(blocks back to the least-recently-active source) block loads → worst case the entire store.

After

```
mset.resetSourceInfo()
var smv StoreMsg
unresolved := map[string]*sourceInfo{}

// Phase 1: targeted index lookup per source.
for _, ssi := range mset.cfg.Sources {
    si := mset.sources[ssi.iname]
    if si == nil { continue }
    var subj string // FilterSubject, or single c
    switch {
    case len(ssi.SubjectTransforms) == 0: subj = ssi.FilterSubject
    case len(ssi.SubjectTransforms) == 1: subj = ssi.SubjectTransforms[0].Destin
    }
    if subj == _EMPTY_ || subj == fwcs || strings.Contains(subj, "{{") {
        unresolved[ssi.iname] = si // can't attribute by a single
        continue
    }
    sm, err := mset.store.LoadLastMsg(subj, &smv) // 0(1) via psim/fss, returns
    if err != nil || sm == nil || len(sm.hdr) == 0 {
        unresolved[ssi.iname] = si
        continue
    }
    if ss := sliceHeader(JSStreamSource, sm.hdr); len(ss) > 0 {
        if _, iName, sseq := streamAndSeq(bytesToString(ss)); iName == ssi.iname {
            si.sseq, si.dseq = sseq, 0 // verified – resolved without
            continue
        }
    }
    unresolved[ssi.iname] = si
}

// Phase 2: only the leftovers pay the (now tiny) reverse scan.
if len(unresolved) > 0 {
    mset.reverseScanForSources(unresolved) // = today's loop, scoped to
}
```

Cost: Phase 1 = $O(S)$ targeted lookups (recent sources share/cache the last block; a quiet source is **one** targeted load, not a walk). Phase 2 only runs for ambiguous sources, with a smaller sublist.

Multi-destination transform sources: call `LoadLastMsg` for each destination and keep the highest origin `sseq`, or defer them to Phase 2. Either is fine; deferring is simplest to start.

6. Before → after (complexity)

	Before	After
Index used	none (generic message walk)	<code>psim</code> / <code>fss</code> per-subject last-block index
Blocks loaded	back to least-active source ($\leq$ all blocks)	~1 per source, only the blocks actually holding a source's last msg
Sensitivity to a quiet source	drags the scan back across the whole store	none — jumps straight to its block
Sensitivity to store size	linear in blocks scanned	none
Worst case still possible?	normal case	only if every source is <code>> /wildcard/shared-subject/pre-2.10</code> (→ Phase 2 == today)
Held lock	<code>mset.mu</code> for the whole walk	<code>mset.mu</code> for O(S) lookups; Phase 2 only for leftovers

The edge → hub topology (each edge mapped to a distinct subject/domain) lands entirely in Phase 1 → the backward scan disappears for that case.

7. Phase 2: keeping the fallback scan narrow

Phase 1 removes most sources from the picture, but whatever remains still runs the reverse `LoadPrevMsgMulti` scan. That scan's whole speed advantage depends on the **sublist being narrow** — `prevMatchingMulti` uses each block's `fss` to skip blocks that contain none of the sublist's subjects. A single `>` (full wildcard) in the sublist defeats that: it matches every subject, so no block can be skipped and the scan visits every block back to the match.

Phase 2 sublist narrowing: one transform source no longer poisons the whole scan

BEFORE — transform/catch-all source injects ">" (empty FilterSubject)



AFTER — phase 1 resolves wildcard transforms; phase 2 sublist uses destinations

step 1: phase 1 collapses wildcard(/)\$N tokens and resolves via the index

tout.{{wildcard(1)}} → tout.* → LoadLastMsg("tout.*") ✓

step 2: only genuinely-unresolved sources remain; sublist stays narrow



Remaining ">" cases (by design): a genuine catch-all source (empty filter, no transform) or an exotic transform (partition/split).

What used to inject >

1. **Every transform source.** A transform source has an empty `FilterSubject` (mutually exclusive with `SubjectTransforms`, `stream.go:1976`), so the old `refreshSublist` did `sl.Insert(fwcs)`. One transform source in the unresolved set widened the whole sublist to `>` → full backward scan.
2. **Templated transforms were deferred to Phase 2 at all.** Their *rendered* destination is a narrow wildcard (e.g. `tout.{{wildcard(1)}}` always lands on `tout.*`), so they don't need a full scan.
3. **Dead entries.** Phase 2 also inserted `si.sfs` — the transform *source* filters (e.g. `tin.*`) — which never match stored subjects (messages are stored under the *destination*).

The two changes

- **Resolve wildcard transforms in Phase 1.** `transformUntokenize(dest)` collapses `wildcard()` / `$N` mapping tokens to a subject wildcard (`tout.{{wildcard(1)}} → tout.*`); `LoadLastMsg` accepts that wildcard and the header- `iname` check still guards attribution. So the common templated transforms resolve in Phase 1 and never reach the scan.
- **Build the Phase 2 sublist from destinations.** For any transform source that *does* reach Phase 2 (e.g. its wildcard space is shared with another source, so Phase 1 saw a header for the other one), insert the destination's wildcard form instead of `>`, and drop the dead `si.sfs` inserts.

What still forces > (by design)

Only a source that *genuinely* needs a broad match:

- a **catch-all source** — empty `FilterSubject`, no transform — sourcing every subject of its origin; and
- an **exotic transform** (`partition` / `split` / `slice` /...) whose rendered destination isn't a subject wildcard, so `transformUntokenize` can't reduce it (it leaves the `{{...}}` token in place, which we detect and fall back to `>`).

Both are uncommon for the edge → hub fan-in. Distinct subjects, wildcard filters, and `wildcard()` / `$N` transforms all either resolve in Phase 1 or keep the Phase 2 sublist narrow enough to skip blocks.

Correctness note: this is purely about *narrowing*, never about *missing* a source. If a narrowed subject is wrong (shared space), the header check fails and the source stays in the unresolved set; an exotic/empty case falls back to `>` . So Phase 2 still finds every source it did before — it just loads far fewer blocks getting there.

8. Correctness & edge cases

- **Same semantics.** Today's scan records, per source, the **most recent** stored message's origin seq (first hit scanning backward). `LoadLastMsg` returns exactly that message. Deleted/interior messages are handled by `loadLast` (it skips `dmap` entries and walks to the previous block if needed), matching the scan's behaviour.
- **Shared subjects / direct publishes.** Resolved safely by the header-iname verification → fall to Phase 2.
- **Wildcard / empty (`>`) filters.** Can't be attributed to one source by subject → Phase 2 (same as today for them).
- **Pre-2.10 headers** (no `iname`, only stream name): verification won't match → Phase 2, which keeps the existing stream-name matching path.
- **memstore.** `MemStore` has the simpler linear `LoadPrevMsgMulti`; `LoadLastMsg` there is also index-light, but memstore streams are small, so Phase 2 cost is negligible. (We can add a memstore `loadLast` fast path later if needed.)
- **setStartingSequenceForSources** (the `STREAM.UPDATE` path, `stream.go:4566`) gets the same Phase 1 treatment for the subset of sources it processes.

9. Risks

- **Index freshness.** `psim` / `fss` may lazily need a recalculation (`lastNeedsUpdate`, `recalculateForSubj`); `loadLast` / `MultiLastSeqs` already handle that, so we inherit

correct behaviour.

- **Behavioural parity.** Phase 1 must produce identical `si.sseq` to the old scan for non-ambiguous sources; this is the core test (below). Keeping Phase 2 byte-for-byte as today bounds the blast radius.
- **Encryption/compression.** Phase 1 still loads the one block holding a source's last message (decrypt/decompress), but only that block — no change in correctness, large reduction in volume.

10. Testing & validation

Prototype status — implemented

A working prototype of the two-phase resolver is implemented in `startingSequenceForSources` (`server/stream.go`), with:

- `TestJetStreamStartingSequenceForSourcesIndexFastPath` (`server/jetstream_sourcing_resume_test.go`) — three distinct-subject sources (index fast path) plus one **subject-transform** source (phase 2 fallback), each with a different origin depth and buried under 20k direct publishes. Asserts every recovered `si.sseq` equals the expected last origin sequence. **Passes.**
- `TestJetStreamStartingSequenceForSourcesAmbiguity` — a catch-all (empty filter) source and **two sources transformed onto the same destination subject** (shared stored subject), mixed with a distinct-subject source. Asserts the phase 2 fallback disambiguates all of them. **Passes.**
- `TestJetStreamSetStartingSequenceForSourcesIndex` — the same fast path applied to the `STREAM.UPDATE` twin `setStartingSequenceForSources` (distinct subject + catch-all + a **concrete-destination transform** source), clearing and recovering `sseq`. **Passes.**
- **Transform fast path:** a source with a single subject transform is resolved in Phase 1 by collapsing `wildcard()` / `$N` mapping tokens to a wildcard form (`transformUntokenize` , e.g. `tout.{{wildcard(1)}}` → `tout.*`) and doing `LoadLastMsg(wildcardForm)` , verified by the header. Only multi-transform and **exotic** transforms (`partition` / `split` / ... that don't reduce to a subject wildcard) still defer to Phase 2.
- **Phase 2 no longer degrades to `>` for transform sources.** Previously a transform source (empty `FilterSubject`) inserted a `>` catch-all into the sublist, defeating per-block skipping for the whole scan. The Phase 2 sublist is now built from the transform **destinations** (wildcard form), with the dead `si.sfs` source-filter inserts removed; only a genuine catch-all source or an exotic transform still forces `>` .
- The change was also applied to `setStartingSequenceForSources` itself (`stream.go:4566`), so both the leader-election and the config-update resume paths use the index fast path and the narrowed sublist.

- Tests (all pass, incl. `-race`): `...IndexFastPath`, `...Ambiguity` (catch-all + shared concrete destination), `...TemplatedTransform` (wildcard transform), `...SharedWildcardTransform` (two templated transforms sharing a `gout.*` space), `TestJetStreamSetStartingSequenceForSourcesIndex` (twin), `TestJetStreamSourcingResumeAfterRolloutRestart` (end-to-end hard-restart, exactly-once resume), `TestJetStreamClusterSourcingResumeAfterLeaderStepDown` (R3 leader-election resume), and `TestJetStreamStartingSequenceForSourcesDifferential` (randomized layouts vs. a brute-force reference).
- `BenchmarkJetStreamScanForSources` (existing, single source), `BenchmarkJetStreamScanForSourcesMulti` (16 sources spread across the store), `BenchmarkJetStreamSourceResumeLeafnodeFanIn` (8–512 edges feeding a hub, quiet edges buried under an all-sourced tail), `...FanInTailDepth` (fixed edges, varying tail depth — isolates the store-depth axis), `...PartitionFallback` (the by-design `>` path), and `...DeepStore` (direct store seeding up to 16/64/256 sources × 10M, for at-scale and cross-design comparison). See the tables below.
- Existing sourcing suite (`SourceBasics`, `SourceRemovalAndReAdd`, `WorkQueueSourceRestart`, `SourceWorkingQueueWithLimit`, `StreamSourceWithoutDuplicateWindow`, `MirrorAndSourcesFilteredConsumers`) still passes.

Measured (filestore, single-server)

Benchmark	Before (reverse scan)	After (phase 1)	Speedup
<code>ScanForSources</code> — 1 source, ~100k buried msgs	~8.5 µs/op	~5.5 µs/op	~1.5×
<code>ScanForSourcesMulti</code> — 16 sources, spread	~110.8 µs/op	~14.0 µs/op	~7.9×

The single-source gap is modest because `LoadPrevMsgMulti` already skips non-matching messages per block; the win scales with the number of sources (each avoids a re-walk via a direct `psim` jump), which is exactly the edge → hub fan-in case.

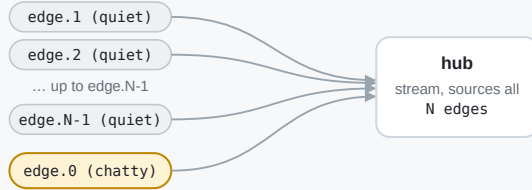
Leafnode fan-in resume (the leader-election case)

`BenchmarkJetStreamSourceResumeLeafnodeFanIn` models the scenario directly: a hub sources from N edge/leafnode streams, then one chatty edge sources a deep 100k tail that buries the other edges' last messages. Crucially the hub store is **entirely sourced edge subjects** — as a real fan-in is — so every block matches the recovery sublist and *none* can be skipped. This is the work a freshly elected leader runs (`startingSequenceForSources`) before it can resume sourcing. One op == one full recovery.

BenchmarkJetStreamSourceResumeLeafnodeFanIn — recovery on leader election

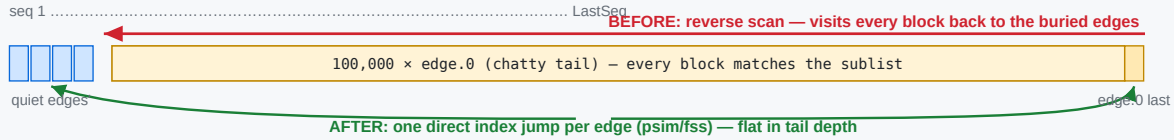
N edge/leafnode streams source into one hub; one chatty edge buries the quiet edges. Measures startingSequenceForSources (one op = one full resume).

1 • Setup — fan-in into the hub



Each quiet edge sends 1 message up front.
The chatty edge then sources a deep tail
tailDepth = 100,000
The hub store is entirely sourced edge
subjects — a real fan-in — so no block in
the tail can be skipped by per-block fss.

2 • Hub store after setup, and how each strategy traverses it



3 • What recovery does

BEFORE — reverse block scan

LoadPrevMsgMulti walks back from LastSeq. The tail is all in-sublist, so nothing is skipped → walks all 100k to reach the buried edges; rebuilds the sublist per source found.
cost ≈ tail depth + O(sources²) sublist rebuild

AFTER — per-subject index (phase 1)

LoadLastMsg(edge.i) jumps straight to each edge's last message via the psim/fss index; header verifies the source. No reverse scan, no sublist built.
cost ≈ N index lookups (flat in tail depth)

4 • Measured (filestore, single-server, benchstat n=6, p=0.002 — geomean -96.1%)

edges	before	after	Δ
8	7.27 ms	0.194 ms	-97.3%
32	7.96 ms	0.286 ms	-96.4%
128	13.2 ms	0.609 ms	-95.4%
512	71.1 ms	3.68 ms	-94.8%

before = store-bound (walks the tail);
after = source-bound (N lookups).
512: O(sources²) rebuild dominates before.

edges	Before (reverse scan)	After (phase 1)	Δ
8	7.27 ms ±96%	0.194 ms ±23%	-97.3%
32	7.96 ms ±6%	0.286 ms ±8%	-96.4%
128	13.2 ms ±6%	0.609 ms ±19%	-95.4%
512	71.1 ms ±22%	3.68 ms ±45%	-94.8%
geomean	15.3 ms	0.594 ms	-96.1%

(filestore, single-server, -benchtime=50x -count=6, benchstat; all rows p=0.002 (n=6). Both columns on the same shared host; "before" measured by splicing in the pre-optimization function from the parent commit. The host is noisy — note the ±96% spread on the edges=8 "before" outlier — but the per-row deltas and the -96.1% geomean are stable across runs.)

Two effects compound, and the table separates them:

- **Store-bound vs source-bound.** "Before" carries a ~10 ms floor independent of edge count — it must reverse-scan the whole 100k tail to reach the buried quiet edges. "After" is flat in tail depth and scales only with edge count (one LoadLastMsg index jump each).
- **The O(sources²) sublist rebuild.** The old Phase 2 rebuilt the whole sublist on every source it resolved (refreshSublist per update). At 512 sources that is ~260k inserts per recovery — the "before" jumps to ~70–100 ms and (in a separate -benchmem run)

49.8 MB / 674k allocs/op. Phase 1 resolves these without ever building a sublist, so allocations drop to **2,572/op** — ~260× fewer at 512 edges.

Caveat: this is a micro-benchmark of the recovery function, not a full Raft election — an actual election adds a fixed quorum/round-trip cost on top of *both* columns equally. An earlier draft buried the edges under *direct* (non-sourced) hub traffic; there the old per-block `fss` skipping rescued the scan (only ~5×), which is not representative of a pure fan-in. The committed benchmark uses an all-sourced tail so nothing is skippable.

Isolating the store-depth axis

`BenchmarkJetStreamSourceResumeFanInTailDepth` fixes the edge count (32) and varies only the buried tail depth. It makes the store-bound vs source-bound split explicit: the old scan grows linearly with depth while Phase 1 stays flat, so the speedup *grows* with how deep the store is.

tail depth	Before (reverse scan)	After (phase 1)	Speedup
50k	0.89 ms	0.30 ms	3.0×
100k	11.6 ms	0.97 ms	11.8×
200k	23.1 ms	1.13 ms	20.5×
400k	46.0 ms	1.11 ms	41.6×

"Before" doubles with the tail (11.6 → 23.1 → 46.0 ms) — textbook $O(\text{depth})$. "After" is flat at ~1 ms and its allocations stay at **~223/op** regardless of depth, versus ~3,400/op for the old scan. (The 50k "before" is sub-linear because at that size the buried edges still sit within the first couple of blocks the scan reaches; from 100k on, the linear walk dominates.)

At scale (10M), and vs. a persisted resume map

A parallel design (see §11) persists a `map[source]→lastSeq` and *reads* it on recovery instead of recomputing from the store. To compare on the same axis,

`BenchmarkJetStreamSourceResumeDeepStore` seeds the hub store directly — real end-to-end sourcing can't build a 10M-message store — with one `JStreamSource`-headered anchor per source plus a deep tail, and a built-in check verifies phase 1 resolves every source before timing. This pushes the reverse-scan baseline to 10M:

sources	tail	Before (reverse scan)	After (index recompute)	Speedup
16	100k	8.6 ms	0.265 ms	33×

sources	tail	Before (reverse scan)	After (index recompute)	Speedup
16	1M	96.9 ms	0.249 ms	389×
16	10M	972 ms	0.265 ms	3,668×
64	10M	974 ms	1.05 ms	928×
256	10M	991 ms	3.08 ms	322×

Both axes confirm the model: the reverse scan is $O(\text{depth})$ ($8.6 \rightarrow 97 \rightarrow 972$ ms across $100k \rightarrow 1M \rightarrow 10M$) and $\sim$ flat in source count (it walks the whole tail regardless — $972/974/991$ ms for $16/64/256$ at $10M$), while the index recompute is flat in depth (~ 0.27 ms at 16 sources from $100k$ to $10M$) and scales with source count (~ 12 μ s/source).

Cross-checking the persisted-map design's published numbers (different host, larger messages):

sources / tail	their scan baseline	their persisted map	our scan baseline	our index recompute
16 / 10M	5.03 s	2.08 μ s	0.972 s	0.265 ms

The two scan baselines have the same $O(\text{depth})$ shape; the $\sim 5\times$ absolute gap is environment (message size / host). The persisted map is $\sim 100\times$ faster than our index recompute *in absolute terms* because it never touches the store — a pure in-memory read — whereas the index still loads one block per source. Those two factors compose to explain why their headline multiplier ($\sim 2.4M\times$) dwarfs ours ($\sim 3,700\times$): $2.42M / 3,668 \approx 660 \approx 5.2$ (baseline gap) $\times 127$ (recompute vs map-read). The designs are **complementary, not competing** — see §11.

What still forces a full `>` scan in Phase 2 (by design)

After the above, Phase 2 only widens to `>` (no block skipping) when a source *genuinely* needs it:

- a **catch-all source** — empty `FilterSubject`, no transform — that sources every subject of its origin (it really can land anywhere, so a broad match is correct); and
- an **exotic transform** (`partition` / `split` / `slice` / ...) whose rendered destination is not a subject wildcard, so `transformUntokenize` can't reduce it to a matchable pattern.

Both are uncommon for the edge $\rightarrow$ hub fan-in. Everything else — distinct subjects, wildcard filters, `wildcard()` / `$N` transforms — either resolves in Phase 1 or narrows the Phase 2 sublist to a concrete or wildcard subject.

`BenchmarkJetStreamSourceResumePartitionFallback` measures this residual path with a `partition()` transform source (stored subjects `p.<n>`, which can't be reduced to a wildcard)

alongside several distinct edges (always resolved in Phase 1), over a deep 100k all-sourced store:

placement of the <code>></code> source	recovery	note
active (it's the chatty aggregator, last msg at LastSeq)	0.65 ms	the common case — the <code>></code> scan resolves in O(1) despite the fallback
buried (quiet early, a distinct edge forms the tail)	56.2 ms	residual worst case — the <code>></code> scan must walk the whole tail

The takeaway: the by-design `>` fallback is cheap whenever the catch-all/partition source is *active* (which is the normal state for an aggregator). It only degrades to a full scan when such a source is both present **and** has gone quiet under a deep tail — genuinely unavoidable, since a catch-all source has no single subject to index by. The distinct edges in the same stream are unaffected either way (Phase 1).

Testing plan & coverage

The recovery path has two layers worth testing separately: the **resolver** (`startingSequenceForSources` producing the right `si.sseq`) and the **end-to-end behaviour** (a real restart/election resumes sourcing exactly-once). Below is the current coverage and the prioritized gaps.

Layer 1 — resolver correctness (unit, calls the resolver directly).  in place: `...`

`IndexFastPath` (phase 1 + transform phase 2), `...Ambiguity` (catch-all + shared concrete destination), `...SetStartingSequenceForSourcesIndex` (update-path twin), `...TemplatedTransform` , `...SharedWildcardTransform` .

Layer 2 — end-to-end recovery. Added:

`TestJetStreamSourcingResumeAfterRolloutRestart` — sources across both phases, hard-restarts the server (rolling-upgrade style), publishes a second batch, and asserts each origin's sourced sequences are exactly `1..N` (no gap = no missed resume; no duplicate = no re-sourced run). `TestJetStreamClusterSourcingResumeAfterLeaderStepDown` — the R3 equivalent: steps down the agg leader so a new node runs the resolver, then asserts the same exactly-once property on the new leader. Both pass under `-race` .

Layer 1.5 — differential / property. Added:

`TestJetStreamStartingSequenceForSourcesDifferential` — randomized layouts (mixed source kinds, random counts incl. zero, randomly interleaved depths) asserting the resolver's per-source `sseq` equals an independent brute-force scan of the same store, over a set of fixed reported seeds.

Prioritized gaps:

Pri	Area	Proposed test(s)	Guards against
P0 ✓	Clustered resume (R3)	...ResumeAfterLeaderStepDown (added)	resolver runs per-node on every election
P0 ✓	Property / differential	...Differential (added)	resolver regressions the hand-written cases miss
P1	Store-state edges (seeded)	seed the store directly (as ... DeepStore does) to build: pre-2.10 headers (empty iname, stream-name match), direct-publish/source subject overlap , and interior deletes/purge before a source's last message	the header- iname disambiguation and loadLast 's dmap /prev-block walk — hard to build via the JS client
P1	memstore path	run the resolver matrix on a MemStore -backed stream	the linear LoadPrevMsgMulti / index-light LoadLastMsg path is untested for sources
P1	Exotic transforms	partition() / split() source resume correctness (the by-design > fallback)	the residual Tier 2 path; perf already benched (...PartitionFallback), correctness not asserted
P2	Config-update path	STREAM.UPDATE adding/removing sources, then assert only affected sources are recomputed and the rest preserved	setStartingSequenceForSources scoping
P2	Snapshot restore	restore a stream from snapshot/backup and assert resume	the cold-restore branch of recovery
P2	First-seq > 1	resume after age/limits expiry has advanced FirstSeq	off-by-one in the reverse-scan termination

Tier 0 (gated on implementation). When the persisted map lands, add a watermark matrix: upToSeq == LastSeq → trust; upToSeq < LastSeq → recompute (Tier 1); upToSeq > LastSeq → distrust; plus checksum-corruption and version-mismatch → fall through (degrade, not error); a crash between message durability and map flush → stale → Tier 1; clean Stop → Tier 0 hit; and a cluster snapshot round-trip.

Still to do

- The **P1** items are the next additions — the seeded store-state edges (pre-2.10 headers, subject overlap, interior deletes) and the memstore path close the remaining hand-

coverage gaps; exotic `partition()` / `split()` resume correctness completes the Tier 2 picture.

Resolved finding (was suspected pre-existing bug)

An earlier note suspected `setStartingSequenceForSources` mishandled transform sources because its Phase 2 sublist used `si.sfs` (transform **source** filters) rather than the **destination**. On inspection this was **not** a correctness bug: transform sources have an empty `FilterSubject`, so the old sublist inserted a `>` catch-all that matched the destination anyway (verified by `TestJetStreamStartingSequenceForSourcesTemplatedTransform`). It was purely an **efficiency** problem — the `>` defeated block skipping — now fixed by building the Phase 2 sublist from the destination wildcard form and dropping the dead `si.sfs` inserts.

11. Relationship to the durable-consumer proposal

This change is **orthogonal and complementary** to the durable-consumer/ `si.sseq` -persistence ideas in `sourcing-durable-resume.md`:

- Durable consumers / persisted `si.sseq` aim to **skip** resume work entirely (when state survives).
- This proposal makes the **fallback resume itself cheap** — which still runs on cold start, on snapshot restore, for limits/clustered streams, and any time persisted state is missing.

The two were measured head-to-head at 16 sources / 10M tail (see "At scale" above): a persisted resume map reads in ~2 μ s (pure in-memory, never touches the store), while the index recompute resolves in ~0.27 ms (one block load per source). Both are flat in store depth — versus ~1 s for the old reverse scan. The persisted map is the faster steady-state path, but it buys that speed with a write-path cost and a **new crash-consistency invariant**: the map must never lead durably-stored messages across truncation, compaction, message deletion, snapshot restore, and follower replication, or recovery resumes at a wrong sequence (gaps/duplicates). The index recompute carries none of that — the log plus the existing subject index remain the only source of truth — so it is the natural path to run whenever the persisted map is absent (pre-feature streams, cold start, restore) or must be revalidated.

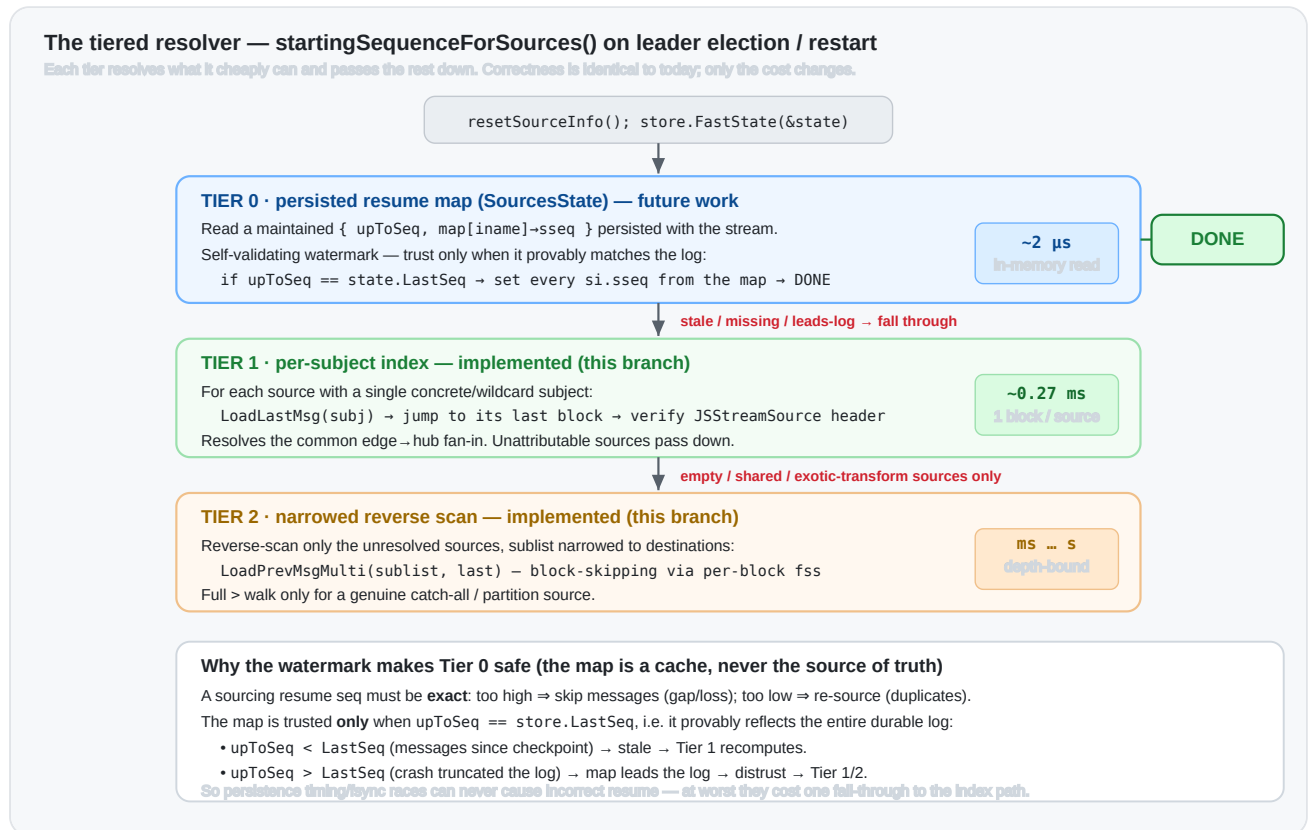
Recommended order: land this index-based resume first (self-contained, no protocol/state change, helps every retention type today, and replaces the $O(\text{depth})$ scan in the fallback the persisted design will still need), then layer the persistence/durable improvements on top.

12. Design sketch: Tier 0, a persisted resume map

This section sketches how the persisted-map approach (a parallel effort) would slot in as **Tier 0** on top of the resolver this branch already implements, what it buys, and the options/trade-offs for *where* the map lives. Nothing here is implemented on this branch — it is the design we'd build the two efforts toward.

12.1 The tiered resolver

`startingSequenceForSources` becomes three tiers, each resolving what it cheaply can and passing the remainder down. Correctness is identical at every tier — only the cost changes.



Tier	Mechanism	Resolves	Cost (16 src / 10M)	Status
0	read a persisted <code>{ upToSeq, map[iname]→sseq }</code>	everything, if the map matches the log	~2 μs	design
1	<code>LoadLastMsg(subj)</code> per source via the subject index	distinct concrete/wildcard-subject sources	~0.27 ms	this branch
2	narrowed <code>LoadPrevMsgMulti</code> reverse scan	residual catch-all / exotic-transform sources	ms ... s	this branch

Tier 0 is a pure *fast path*: when present and trustworthy it returns immediately; otherwise the work falls through to the index recompute (Tier 1) and, for the genuine residue, the narrowed scan (Tier

2). Because Tiers 1–2 already exist and are cheap, Tier 0 can be added incrementally with no correctness risk to the fallback.

12.2 How Tier 0 works

The per-source resume sequence (`si.sseq` — the origin sequence of the last message sourced from each source) is already tracked in memory: it is set in `processInboundSourceMsg` (`stream.go`) as each sourced message is ingested, and today it is **always recomputed** from the store on leader election (`setLeader` → `setupSourceConsumers` → `startingSequenceForSources`). No source/mirror resume state is persisted anywhere today.

Tier 0 persists that in-memory map so recovery can *read* it instead of recomputing:

1. **Maintain** `map[iname]→sseq` — already done; it is the set of `si.sseq` values.
2. **Persist** the map, tagged with a watermark `upToSeq` = the hub stream `LastSeq` the map reflects.
3. **On recovery**, load the map and decide whether to trust it (§12.4); on a hit, set every `si.sseq` from the map and return — no store access at all.

12.3 The consistency requirement (why a naive map is unsafe)

A sourcing resume sequence must be **exact**, not merely "not ahead":

- **Map too high** (claims a seq the store doesn't durably have — e.g. a crash truncated messages written after the last checkpoint) ⇒ we resume *past* real messages ⇒ **gap / data loss**.
- **Map too low** (stale — messages were sourced since the last checkpoint) ⇒ we re-request from the origin and re-append ⇒ **duplicates** (sourced messages carry no `Nats-Msg-Id` , so the store's dedupe does not catch them).

So "persist it and trust it" is only safe if the map is *exactly* the store's state — which is precisely the crash-consistency burden a persisted map introduces, and why the state is recomputed today.

12.4 The watermark: self-validating trust

The cheap way to get exactness without per-source verification (which would just re-do Tier 1) is a single **watermark**. Persist `upToSeq` alongside the map and, on recovery, compare it to the store's actual `LastSeq` :

Condition	Meaning	Action
<code>upToSeq == LastSeq</code>	map provably reflects the entire durable log	trust it — Tier 0 hit (~µs)

Condition	Meaning	Action
<code>upToSeq < LastSeq</code>	messages were stored after the checkpoint	stale → fall to Tier 1
<code>upToSeq > LastSeq</code>	crash truncated the log below the checkpoint	map leads the log → distrust → Tier 1/2

This makes Tier 0 **safe by construction and independent of fsync timing**: if the last few sourced messages weren't durable and are lost on restart, the recovered `LastSeq` is below `upToSeq`, the map is distrusted, and Tier 1 recomputes. The worst case of *any* persistence race is one fall-through to the index path — never an incorrect resume. The common clean-shutdown / steady-checkpoint case (the typical restart and rolling-upgrade path) hits `upToSeq == LastSeq` and pays $\sim\mu\text{s}$.

12.5 Where to persist the map — options

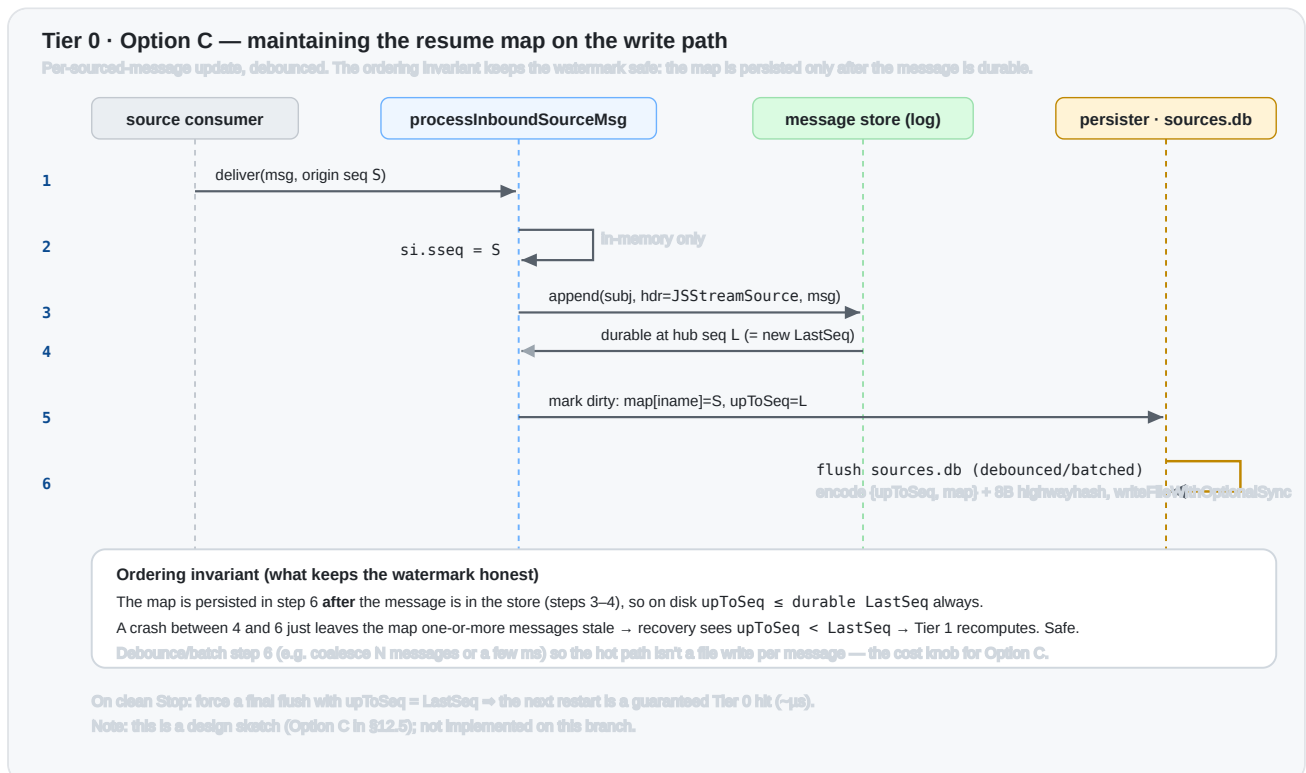
The map is tiny (`iname` → `uint64` , a few dozen entries), so the question is purely *where* it is written to stay consistent and *how often*. Five options, roughly increasing in consistency strength and in write-path / conflict cost:

Option	Mechanism	Pros	Cons
A. Inline in <code>index.db</code>	encode the map inside the filestore full-state file (<code>_writeFullState</code> / <code>recoverFullState</code>)	atomic with stream state; one file; watermark = the state's <code>LastSeq</code> for free	touches the checksummed filestore format (highest conflict surface); only as fresh as the periodic full-state write
B. Sidecar <code>sources.db</code>	a small separate file in the stream's <code>msgs/</code> dir, mirroring the consumer-state file pattern (<code>encodeConsumerState</code> / <code>writeState</code> / highwayhash sum)	self-contained; no change to <code>index.db</code> ; easy to version & checksum	a second file to write/sync and keep in step; its own staleness window
C. Write-path update	update + persist the map on every sourced message (the "always $\sim 2\ \mu\text{s}$ on recovery" design)	map is never stale (<code>upToSeq</code> $\approx$ <code>LastSeq</code> almost always) ⇒ Tier 0 hit even after a crash	per-message write amplification; needs batching/debounce; strongest crash-consistency coupling

Option	Mechanism	Pros	Cons
D. Periodic + on-stop	snapshot <code>{LastSeq, map}</code> on a timer and on clean <code>Stop</code>	trivial; no write-path cost; on-stop write makes clean restarts a guaranteed Tier 0 hit	after a <i>crash</i> , almost always stale $\Rightarrow$ falls to Tier 1 (still ~ 0.27 ms, so acceptable)
E. RAFT snapshot (clustered)	add the map to the stream's replicated snapshot encoding (<code>StreamReplicatedState</code> / <code>stateSnapshot</code>)	replicated to followers; survives leader change without a recompute	changes the replicated state format & version; must stay consistent across catchup/restore

Notes:

- **A vs B vs D differ only in the staleness window**, and the watermark makes any window *safe* — it only affects the Tier 0 *hit rate*, not correctness. So a low-risk first cut is **D** (periodic + on-stop, sidecar file), upgrading to **C** later if steady-state recovery latency on crash matters.
- **C** is what yields the headline " $\sim 2 \mu$ s even after a crash," at the cost of touching the hot write path; it is the most invasive and the one most worth measuring for write-amplification. Its update sequence and the ordering invariant that keeps the watermark honest are shown below:



- **E** is orthogonal to A–D and is required for the *clustered* win — otherwise a clustered stream still recomputes (now via Tier 1) when a new leader has no local map. This is the largest piece and the one with the most format/version care.
- Whatever the choice, the map must be **bounded by LastSeq on read** (§12.4) and treated as a hint the log can always override.

12.6 Clustered streams

For replicated streams the resume state lives behind RAFT. Two sub-cases:

- **Same node re-elected / restarted** — a local sidecar/ `index.db` map (A/B/D) applies as in the single-server case.
- **Different node becomes leader** — it may have no local map (or a stale one). Either replicate the map in the stream snapshot (E), or simply let Tier 1 recompute on that node — which, post-this-branch, is ~0.27 ms rather than the old $O(\text{depth})$ scan. That fallback is what makes Tier 0 optional per-node rather than a hard dependency.

12.7 Expected performance

From the measured numbers (§10, "At scale"), with 16 sources over a 10M-message store:

Path	Recovery	Scales with	Flat in store depth?
Old reverse scan (pre-branch)	~0.97 s	store depth	no — $O(\text{depth})$
Tier 1 index recompute (this branch)	~0.27 ms	source count (~12 $\mu\text{s}/\text{src}$)	yes
Tier 0 persisted map (design)	~2 μs (reported)	source count (~0.13 $\mu\text{s}/\text{src}$)	yes

So Tier 0 is **~100× faster than Tier 1 in absolute terms** (a pure in-memory read vs one block load per source) and both are flat in depth. The practical question is whether ~0.27 ms recovery is already good enough: for most deployments it is, and Tier 1 alone removes the multi-second cliff. Tier 0 earns its keep when recovery happens *often* or with *many* sources — frequent leader elections / rolling restarts, or hundreds–thousands of sources where 0.27 ms → tens of ms (Tier 1) is worth driving back to μs . It does **not** change the asymptotics (both are $O(\text{sources})$); it lowers the constant by removing store I/O.

12.8 Recommendation

1. **Ship Tiers 1–2 now** (this branch): self-contained, no new state, removes the $O(\text{depth})$ cliff for every retention type, and is the fallback Tier 0 needs anyway.

2. **Add Tier 0 incrementally:** start with option **D** (periodic + on-stop sidecar, watermark-validated) for single-server/R1 — small, safe, and already turns clean restarts into μ s. Measure the write path before considering **C**.
3. **Then Tier 0 for clusters** (option **E**): the largest and most consistency-sensitive piece; gate it behind the same watermark and keep Tier 1 as the per-node fallback so it can never resume incorrectly.

12.9 Concrete encoding (SourcesState)

The map is small, so a flat varint encoding mirroring `encodeConsumerState ( store.go:401 )` is plenty. It reuses the filestore conventions: a 2-byte header (`magic = 22` , `version = 1` , `hdrLen = 2` , `filestore.go:288–294`) and an 8-byte trailing highwayhash-64 checksum over the preceding bytes (`checksumSize = 8` , computed with the store's `fs.hh` digest as in `_writeFullState`).

Byte layout:

Offset	Field	Type	Notes
0	<code>magic</code>	<code>u8</code>	<code>= 22</code> (filestore magic)
1	<code>version</code>	<code>u8</code>	<code>= 1</code>
2...	<code>upToSeq</code>	<code>uvarint</code>	hub <code>LastSeq</code> the map reflects — the watermark (§12.4)
...	<code>n</code>	<code>uvarint</code>	number of source entries
...	<code>len(iname)</code>	<code>uvarint</code>	↵ repeated <code>n</code> times
...	<code>iname</code>	<code>bytes</code>	the source's unique index name
...	<code>sseq</code>	<code>uvarint</code>	└ last origin sequence sourced for it
<code>end-8</code>	<code>checksum</code>	<code>[8]byte</code>	highwayhash-64 over <code>buf[0 : end-8]</code>

Encode (sketch):

```
const sourcesStateMagic = magic // 22, shared filestore magic
const sourcesStateVersion = uint8(1)

// encodeSourcesState serializes {upToSeq, map[iname]→sseq}. hh is the store's
// keyed highwayhash digest (fs.hh); pass nil to skip the checksum (tests).
func encodeSourcesState(upToSeq uint64, seqs map[string]uint64, hh hash.Hash64)
    // Upper bound: header + upToSeq + count + per-entry(len + iname + sseq
    sz := hdrLen + 2*binary.MaxVarintLen64
```

```

    for iname := range seqs {
        sz += binary.MaxVarintLen64 + len(iname) + binary.MaxVarintLen
    }
    sz += checksumSize
    buf := make([]byte, sz)

    buf[0], buf[1] = sourcesStateMagic, sourcesStateVersion
    n := hdrLen
    n += binary.PutUvarint(buf[n:], upToSeq)
    n += binary.PutUvarint(buf[n:], uint64(len(seqs)))
    for iname, sseq := range seqs {
        n += binary.PutUvarint(buf[n:], uint64(len(iname)))
        n += copy(buf[n:], iname)
        n += binary.PutUvarint(buf[n:], sseq)
    }
    if hh != nil {
        hh.Reset()
        hh.Write(buf[:n])
        n += copy(buf[n:], hh.Sum(nil)) // 8 bytes
    }
    return buf[:n]
}

```

Decode (sketch) — rejects a bad magic/version/checksum and returns the map plus the watermark, which `startingSequenceForSources` (Tier 0) then compares against `state.LastSeq`:

```

func decodeSourcesState(buf []byte, hh hash.Hash64) (upToSeq uint64, seqs map[string]uint64, err error) {
    if len(buf) < hdrLen+checksumSize || buf[0] != sourcesStateMagic || buf[1] != sourcesStateVersion {
        return 0, nil, errBadSourcesState
    }
    body, sum := buf[:len(buf)-checksumSize], buf[len(buf)-checksumSize:]
    if hh != nil {
        hh.Reset()
        hh.Write(body)
        if !bytes.Equal(hh.Sum(nil), sum) {
            return 0, nil, errBadSourcesState
        }
    }
    n := hdrLen
    upToSeq, c := binary.Uvarint(body[n:]); n += c
}

```

```

    cnt, c := binary.Uvarint(body[n:]); n += c
    seqs = make(map[string]uint64, cnt)
    for i := uint64(0); i < cnt; i++ {
        l, c := binary.Uvarint(body[n:]); n += c
        iname := string(body[n : n+int(l)]); n += int(l)
        sseq, c := binary.Uvarint(body[n:]); n += c
        seqs[iname] = sseq
    }
    return upToSeq, seqs, nil
}

```

Size: ~len(iname)+~12 bytes per source, so a 256-source stream is well under 10 KB – a single small write, well within one filesystem block for typical fan-ins. Forward compatibility is the usual version bump (Tier 0 simply distrusts an unrecognized version and falls through to Tier 1, so an old/new format mismatch degrades to the index recompute rather than an error).

The Tier 0 read in `startingSequenceForSources` is then small:

```

// Tier 0: trust a persisted map only if it provably matches the durable log.
if buf, _ := mset.readSourcesState(); len(buf) > 0 {
    if upToSeq, seqs, err := decodeSourcesState(buf, mset.store.hh()); err == nil {
        for iname, sseq := range seqs {
            if si := mset.sources[iname]; si != nil {
                si.sseq, si.dseq = sseq, 0
            }
        }
        return // sources not in the map have never sourced → sseq stays 0
    }
}
// else fall through to Tier 1 (index recompute) / Tier 2 (narrowed scan)

```

Appendix — key references

Symbol	File:line	Role
<code>startingSequenceForSources</code>	<code>stream.go:4694</code>	the scan to replace (Phase 1 + Phase 2)

Symbol	File:line	Role
unconditional call	<code>stream.go:4821</code>	runs on every leader election / restart
<code>setStartingSequenceForSources</code>	<code>stream.go:4566</code>	update-path twin, same treatment
sublist build (stored subjects)	<code>stream.go:4742-4757</code>	defines a source's stored subject(s)
<code>LoadLastMsg</code> / <code>loadLast</code>	<code>filestore.go:9048</code> / <code>8939</code>	index-based last-msg-for-subject (returns header)
<code>MultiLastSeqs</code>	<code>filestore.go:3806</code>	batched last-seq-per-filter via index
<code>psi</code> / <code>fs.psim</code>	<code>filestore.go:168</code> / <code>195</code>	per-subject <code>{total, fblk, lblk}</code> ; store-wide subject → blocks index
<code>mb.fss</code> / <code>SimpleState</code>	<code>filestore.go:239</code> / <code>store.go:180</code>	per-block subject → <code>{Msgs, First, Last}</code> index
<code>psim</code> maintenance on write	<code>filestore.go:4933-4943</code>	how <code>total</code> / <code>lblk</code> are kept current
<code>LoadPrevMsgMulti</code>	<code>filestore.go:9403</code> / <code>memstore.go:1991</code>	the backward walk used by Phase 2
<code>Nats-Stream-Source</code> header	<code>stream.go:635</code>	constant; the per-message source provenance
<code>genSourceHeader</code> / <code>streamAndSeq</code>	<code>stream.go:4470</code> / <code>4545</code>	writes / parses origin stream, iname, origin seq

Tier 0 (design) persistence hooks:

Symbol	File:line	Role
<code>processInboundSourceMsg</code> (<code>si.sseq = sseq</code>)	<code>stream.go:4371</code>	where the in-memory resume map is updated per sourced message
<code>setLeader</code> → <code>subscribeToStream</code> → <code>setupSourceConsumers</code>	<code>stream.go:1255</code> / <code>4957</code> / <code>4927</code>	the recovery trigger Tier 0 would short-circuit

Symbol	File:line	Role
<code>_writeFullState</code> / <code>recoverFullState</code>	<code>filestore.go:11681</code> / <code>1871</code>	filestore full-state write/read (<code>index.db</code>) — option A host
<code>encodeConsumerState</code> / <code>writeState</code>	<code>store.go:401</code> / <code>filestore.go:13164</code>	consumer-state file pattern to mirror for a sidecar <code>sources.db</code> (option B)
<code>StreamReplicatedState</code> / <code>stateSnapshot</code>	<code>store.go:229</code> / <code>jetstream_cluster.go:9897</code>	replicated stream state / snapshot — option E host (no source state today)